BIG IDEAS MATH • GRADE 6 • CHAPTER 3

Algebraic Expressions & Properties: Order of Operations

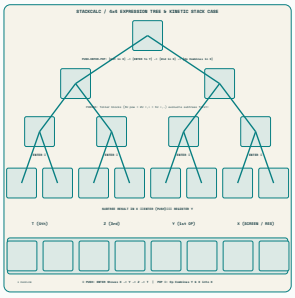
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.6.EE.A.2

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Tree Board

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place operator tokens [+] and [×] at tree junctions; place number tokens at leaf nodes.

Sketch Model on Paper:
Draw an expression tree structure showing operation priority levels.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	4	X: 4	Operand 4 into X
2.	ENTER	Y: 4 | X: 4	Stack lift
3.	5	Y: 4 | X: 5	Operand 5 into X
4.	×	X: 20	Multiply first: 4 ×

EXPECTED LCD X: 23

1 **Evaluate 10 + 6 × 3 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Evaluate 50 - 4 × 8 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Evaluate (8 + 2) × (9 - 4) on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Evaluate 2^3 + 5 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 3

Algebraic Expressions & Properties: Order of Operations • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: 4x4 Expression Tree Case & Token Operators

CAD Blueprint: tree_board

Target Device: SC-6 / SC-32 Bridge

CAD / STL Source File:

Education/build/scad/tree_board.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Expression Tray & White Tokens (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Evaluate 10 + 6 × 3 on the stack.** Standard: CCSS.MATH.6.EE.A.2	28	[6] [ENTER] [3] [×] [1][0] [+]
2	**Evaluate 50 - 4 × 8 on the stack.** Standard: CCSS.MATH.6.EE.A.2	18	[4] [ENTER] [8] [×] [5][0] [SWAP] [-]
3	**Evaluate (8 + 2) × (9 - 4) on the stack.** Standard: CCSS.MATH.6.EE.A.2	50	[8] [ENTER] [2] [+] [9] [ENTER] [4] [-
4	**Evaluate 2^3 + 5 on the stack.** Standard: CCSS.MATH.6.EE.A.2	13	[2] [ENTER] [3] [y^x] [5] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Students often evaluate left-to-right (3+4=7, 7×5=35), ignoring multiplication precedence.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.